OMNISIGHT

PROFESSIONAL LAN MONITORING & CONTROL

TECHNICAL OPERATIONAL MANUAL

VERSION 1.0.0 | MAY 2026

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1. PROJECT OVERVIEW

OmniSight is a high-performance, LAN-based monitoring and control system designed for parental oversight and commercial environments such as Internet Cafes. It provides real-time visibility and administrative command capabilities over local network machines.

KEY FEATURES

- Live Screen Stream: Low-latency JPEG-compressed desktop monitoring.
- Stealth Operation: Client runs as a hidden LocalSystem Windows Service.
- Remote Commands: Instant Lock, Shutdown, and Broadcast Messaging.
- Secure Access: Password-protected Admin Dashboard and client uninstallation.
- Web-Based UI: Modern React/Next.js interface hosted via .NET WebView2.

2. SYSTEM ARCHITECTURE

The system utilizes a client-server model optimized for local network stability.

COMPONENTS

1. Admin Dashboard (.NET + React): The central hub hosting the SignalR server and UI.
2. Managed Client (.NET Worker): The headless service performing capture and execution.
3. SignalR Communication: Real-time duplex communication for commands and data.

3. SETUP & INSTALLATION

ADMIN SETUP

1. Deploy 'installer-admin.exe' on the monitoring PC.
2. Configure Administrator credentials during the first launch.
3. Ensure Firewall ports for SignalR (Default: 5000) are open.

CLIENT DEPLOYMENT

1. Deploy 'installer-client.exe' on target machines.
2. Enter the Admin PC's IP address when prompted.
3. Set a secure uninstallation password to prevent tampering.

4. ADMIN DASHBOARD OPERATION

MONITORING & OPTIMIZATION

Select a device from the 'Active Devices' sidebar to view its live screen. The system is optimized for LAN efficiency: only the currently viewed machine transmits data. Toggling the monitor to 'Offline' or switching devices immediately stops the data stream from the client.

REMOTE COMMANDS

- Broadcast: Send instant pop-up messages to one or all clients.
- System Lock: Immediately restrict user access to the desktop.
- Shutdown: Remotely power off machines for maintenance or closing time.

5. TECHNICAL SPECIFICATIONS

Capture Engine: GDI+ Screen Scraper

Compression: Adaptive JPEG (1-100 Quality)

Transport: SignalR over WebSockets/Long Polling

UI Framework: Tailwind CSS + Next.js (Static Export)

Host: .NET 10.0 WPF with WebView2 Runtime

SECURITY NOTICE

Access to the OmniSight Dashboard is restricted to authorized personnel. Unauthorized use or distribution of this manual or the software is prohibited.